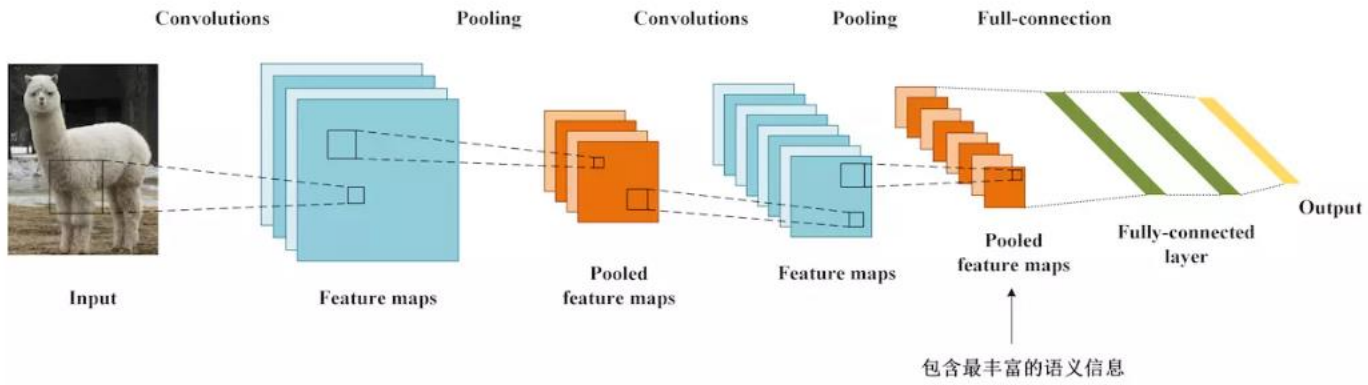
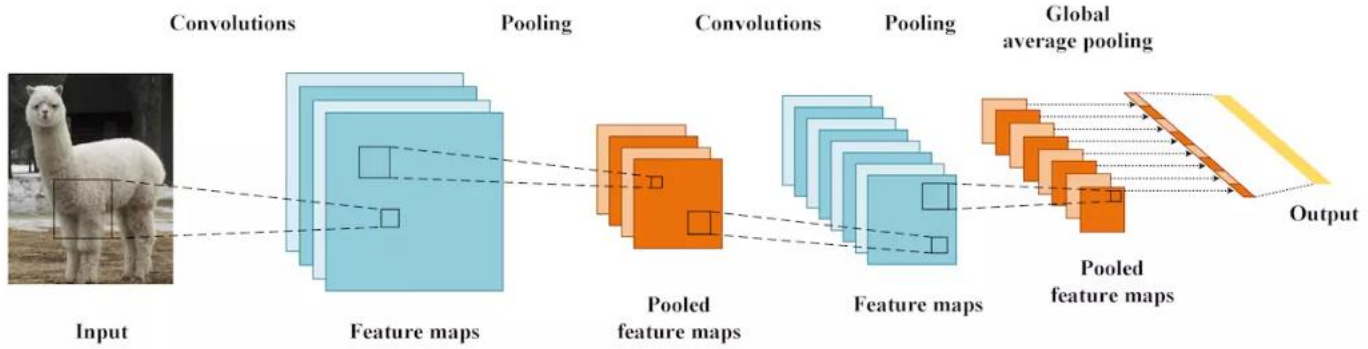
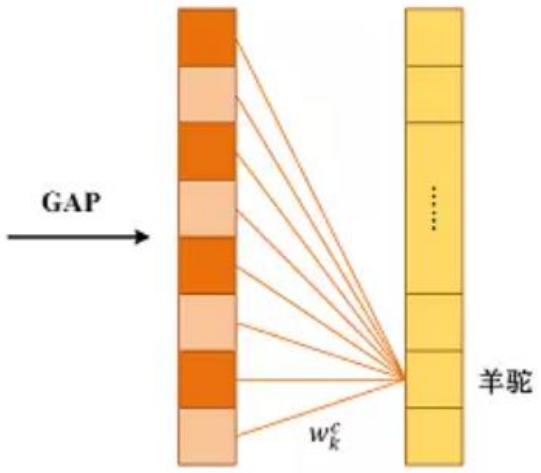
CAM



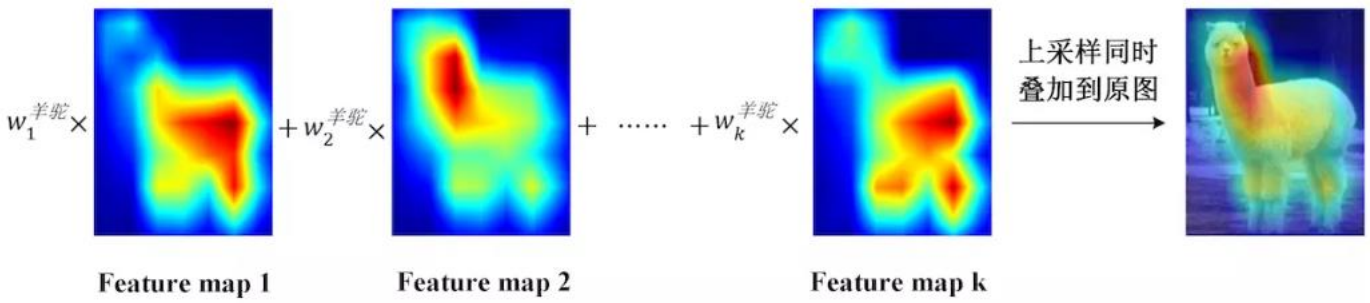
利用GAP(Global Average Pooling)替换掉了全连接层。可以把GAP视为一个特殊的average pool层，只不过其pool size和整个特征图一样大，其实说白了就是求每张特征图所有像素的均值。



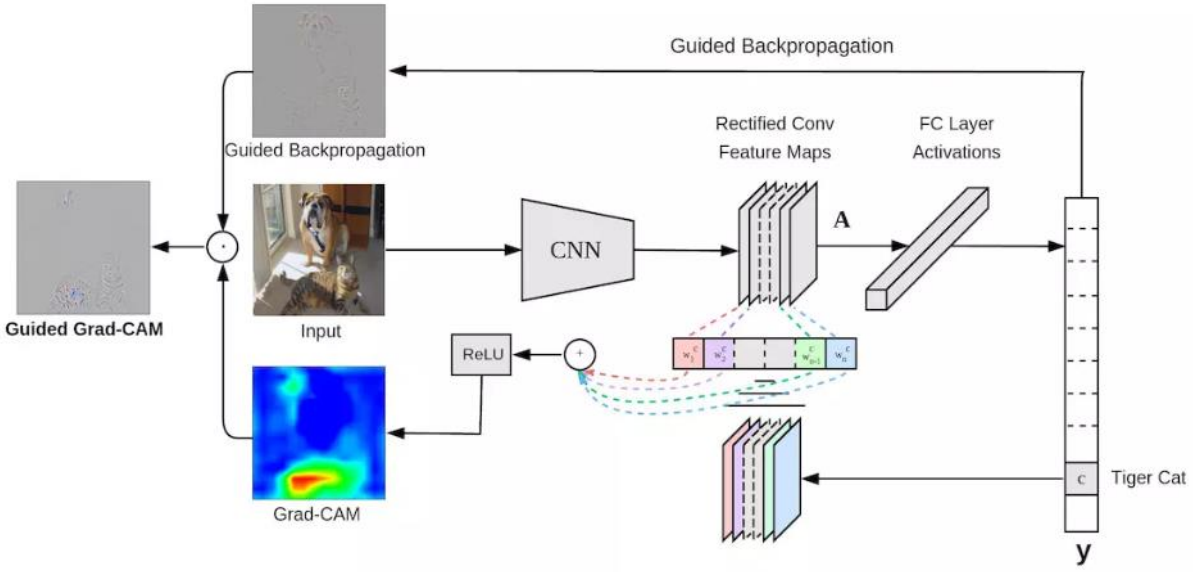
我们重点看下经过GAP之后与输出层的连接关系(暂不考虑softmax层)，实质上也是就是个全连接层，只不过没有了偏置项，如图所示



w为权重，k表示对应的特征图，c为对应的类别



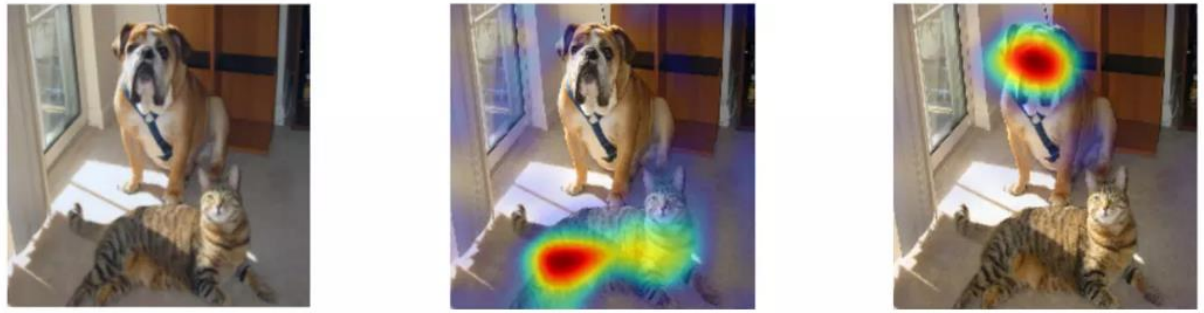
Grad-CAM



使用Grad-CAM 对卷积网络中的特征图进行加权求和，得到卷积conv5的热力图，这种可视化机制必须有一个前置条件就是告诉算法具体的类别，通过这个输出y得到conv5的梯度，对梯度进行平均求和等降维操作，得到conv5中每个通道权重，这样 权重 x conv5 = 热力图，

$$\alpha_k^c = \overbrace{\frac{1}{Z}\sum_i\sum_j}^{\text{global average pooling}} \underbrace{\frac{\partial y^c}{\partial A_{ij}^k}}_{\text{gradients via backprop}} \qquad (1)$$

$$L_{\text{Grad-CAM}}^c = ReLU\underbrace{\left(\sum_k \alpha_k^c A^k\right)}_{\text{linear combination}} \qquad (2)$$

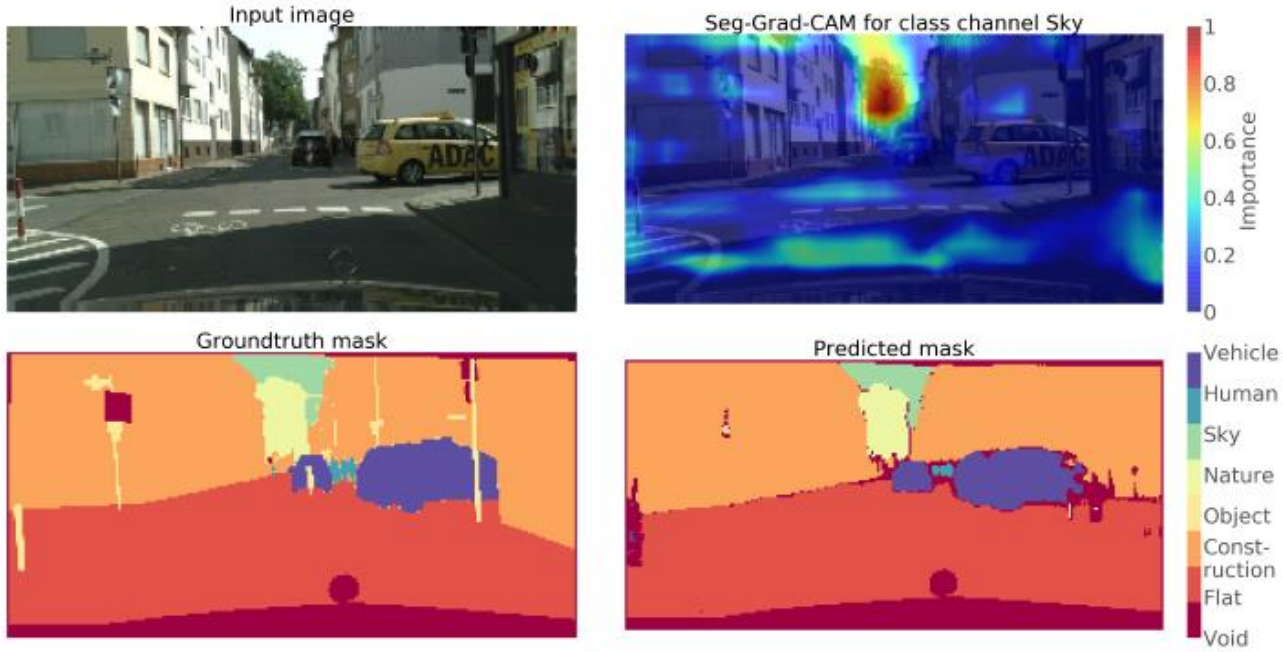


原图　　分类为猫的依据　　分类为狗的依据

SEG-GRAD-CAM

$$L^c = \mathrm{ReLU}\left(\sum_k \alpha_c^k A^k\right) \text{ with } \alpha_c^k = \frac{1}{N}\sum_{u,v}\frac{\partial y^c}{\partial A_{uv}^k} \quad (1)$$

Hence, we propose SEG-GRAD-CAM by replacing y^c by $\sum_{(i,j)\in\mathcal{M}} y_{ij}^c$ in Eq. (1), where $\mathcal{M}$ is a set of pixel indices of interest in the output mask.



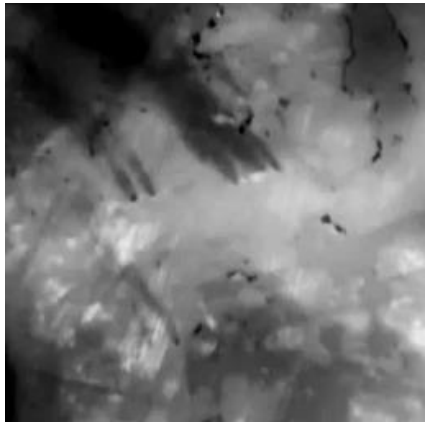
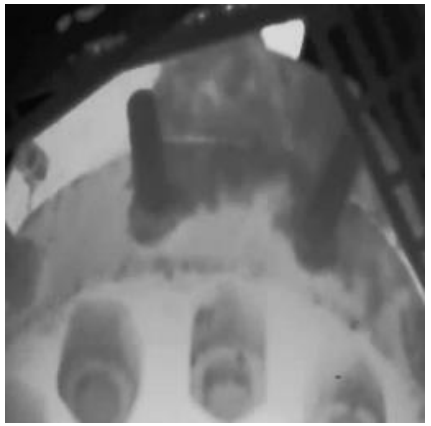
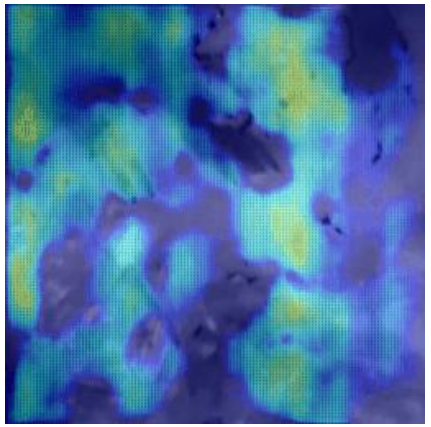
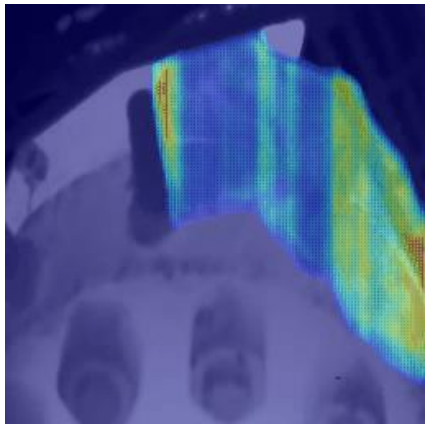
我的

1.前面所有的分类或者分割都是其网络的输出都是one-hot编码（取你想要类别的index）。我的网络层最后一层是sigmoid，所以不是onehot编码，我取的是>0.5的index。
DSS可视化

适用于任何分割网络的注意力图可视化

SEG-GRAD-CAM 3D

改成了基于3DUnet的，也就是网络的数日多了一维。

Hence, we propose SEG-GRAD-CAM by replacing y^c by $\sum_{(i,j)\in\mathcal{M}} y^c_{ij}$ in Eq. (1), where $\mathcal{M}$ is a set of pixel indices of interest in the output mask.

$$\sum_{(i,j,p)} \in y^c_{ij_P}$$